Frontiers in Smart Healthy Ageing (AIHA 8122)

Test 1

Closed-book written examination on foundational theory

Date: Nov 6, 2026

Name **Student ID**

Q1. Evaluating a home-based fall-prevention programme fairly

Scenario: A research team recruits community-dwelling older adults and gives them wearable sensors (step count, sleep, gait). The goal is to estimate whether a home-based exercise programme reduces 90-day fall risk. Adherence to the programme and frailty status change over time, and both affect whether a participant keeps exercising and their subsequent fall risk.

Background hints:

- Time-varying confounding: a variable that changes over time, is affected by past treatment, and also influences both future treatment and the outcome.
- Target trial emulation: designing an observational analysis as if it were a randomised trial — defining eligibility, treatment strategies, a clear start of follow-up ("time zero"), and the outcome in advance.
- Immortal time bias: bias created when participants must survive (stay event-free) for a period before they can be classified as "treated", and that survival time is incorrectly credited to the treatment.

Question (30 points)

(a) Explain in plain language why comparing fall rates between participants who did and did not exercise can give a misleading answer, even with a large sample.

(b) Define "time zero" for this study and explain why choosing it wrongly creates immortal time bias.

Q2. Building an end-to-end fall-risk prediction model

Scenario: You must build a machine-learning model that predicts, at a given "prediction time", whether an older adult will fall within the next 90 days. Data come from electronic health records and a wrist-worn wearable. About 5% of the cohort falls. The wearable has large gaps (often removed during illness), and some variables (e.g., "fall severity", "post-fall mobility score") are only recorded after a fall has occurred.

Background hints:

- Information leakage: the model uses information that would not genuinely be available at prediction time, or information crosses between training and test data.
- Class imbalance: the event of interest (falls) is much rarer than non-events.
- Missing-data mechanisms: MCAR (missing completely at random), MAR (missing at random given observed data), MNAR (missingness itself depends on the unobserved value / health state).

Question (40 points)

(a) Give three concrete examples of information leakage in this scenario and explain why each inflates the estimated performance.

(b) Explain why 95% accuracy would not prove the model is useful, name one better metric, and describe one strategy for handling the class imbalance.

Q3. Should the clinic offer a "biological age" score?

Scenario: A clinic proposes offering a "biological age" score, built from epigenetic, clinical, physiological, and multi-omics data, to guide personalised prevention for older adults.

Background hints

- Chronological age = time since birth; biological age = the body's physiological state.
- Healthspan = the period of life lived with capacity and participation, before disability dominates.
- The 12 hallmarks of ageing form an interacting network (primary -> antagonistic -> integrative), not a single clock.

Question (40 points)

(a) Using the example of two 70-year-olds with different risk profiles, explain what a biological-age score adds beyond chronological age, and why healthspan (not lifespan) is the right target.

(b) List the four data components of the score and explain why you would validate it against a functional outcome (e.g., frailty or disability-free survival) rather than mortality alone.